PROB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Uniform 10-Sector Trial

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why doesn\'t each number land exactly 5 times in 50 spins?'
- Ask: 'What makes the 36° sectors geometrically fair?'
- Ask: 'How does the captive retention barb keep the arrow from wobbling off course?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking a number is 'due' to hit if it hasn't appeared yet (Gambler's Fallacy). Remind them: the spi
nner has no memory!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"What is the theoretical probability of spinning a prime number (2, 3, 5, 7) on the 1–10 dial?"
""

OFFICIAL SOLUTION KEY:

Four primes out of 10 numbers: P(Prime) = 4/10 = 2/5 = 40%.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/dice-and-spinner/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PROB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Uniform 10-Sector Trial

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Is a spinner truly random? In this lab, test the low-friction captive spindle over 50 spins and compare your experimental results to the mathematical truth.

STEP-BY-STEP APPARATUS SETUP:

1. Snap the 10-Sector Uniform Dial onto the spinner base.
2. Give the arrow a test flick: Notice it spins freely on its conical bearing for >5 seconds.
3. Prepare a tally chart for numbers 1 through 10.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Modular Spinner Base + 10-Sector Uniform Dial (36° each, 1–10) + Captive Spindle
DATUM TARGET:	Fairness Invariant: Each Sector Sweeps Exactly 36.0° (P(n) = 1/10 = 0.1000 = 10%)
CALCULATOR KEY:	[1] [/] [10] -> 0.10 (10% per sector)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 6B, Chapter 13 (Theoretical Probability)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7 (Probability & Fair Games)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 10, Sec. 10.1 • CCSS.MATH.7.SP.C.5 • TEKS 7.6I
GRADE LEVEL BAND:	Grades 5–7 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7 ('Data & Probability'), pp. 158–166
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 2 (Fairness Geometry) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE 50-SPIN EXPERIMENTAL RUN

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Spin the arrow 50 times. Record each landing number in your tally chart.

Step 2: Calculate theoretical probability: 1 out of 10 = 10% (expected 5 landings per number in 50 spins).

Step 3: Compare your actual results: Did any number land exactly 5 times? Did any land 7 or 8 times?

Step 4: Conclude: In a small sample (50 spins), experimental results fluctuate around the theoretical average.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Fairness Invariant: Each Sector Sweeps Exactly 36.0° (P(n) = 1/10 = 0.1000 = 10%)

PHYSICAL APPARATUS MANIFEST:

Modular Spinner Base + 10-Sector Uniform Dial (36° each, 1–10) + Captive Spindle

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: CALCULATING RELATIVE FREQUENCY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Calculate the relative frequency of spinning an even number (2, 4, 6, 8, 10).

FORM A MASTER STATION TASKS (Complete All 3):

1 **What is the theoretical probability of spinning an EVEN number?**
Expected Physical Finding: **5/10 = 1/2 = 50%**

2 **Count your team's total even landings out of 50 spins.**
Expected Physical Finding: **Typically 22 to 28 landings (44% to 56%)**

3 **What happens if you combine data with all other tables in the room (300 spins)?**
Expected Physical Finding: **Relative frequency moves much closer to exactly 50%!**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: **[1] [/] [10] -> 0.10 (10% per sector)**

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Target outcome count
10	Y: 1 | X: 10	Total equal sector sample space
/	X: 0.10	Theoretical probability evaluation

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 4: Weighted Realities: The Biased Dial

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does Zone A cover half the circle?'
- Ask: 'How do casinos design games where you win small prizes often but lose money over time?'
- Ask: 'What would the angle of Zone D need to be to give a 1-in-3 chance?' (120°).

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 4 sectors means each has a 1/4 chance. Point directly to the physical sizes of the wedges!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If a new sector E has an angle of 72°, what percentage of the board does it occupy?"
""

OFFICIAL SOLUTION KEY:

72° / 360° = 1/5 = 20%.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

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Point any phone, iPad, or Chromebook camera
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

PROB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 1: Weighted Realities: The Biased Dial

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Not all events in life have equal odds. Weather forecasts, carnival games, and insurance rates use weighted probabilities. In this lab, test an unfair spinner.

STEP-BY-STEP APPARATUS SETUP:

1. Flip the dial to Side B (the Non-Uniform Weighted Dial).
2. Notice the four colored zones: Zone A (Blue, 180°), Zone B (Green, 90°), Zone C (Amber, 54°), Zone D (Red, 36°).
3. Prepare to record 40 spins.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Modular Spinner Base + Side B Weighted Dial (Sector A: 180°, B: 90°, C: 54°, D: 36°)

DATUM TARGET: Angular Invariant: Sector Angle / 360° = Theoretical Probability (180° = 50%, 90° = 25%)

CALCULATOR KEY: [180] [/] [360] -> 0.50 (50% Zone)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7B, Chapter 9 (Experimental Probability)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7, Module 2 (Unfair Spinners)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 10, Sec. 10.2 • CCSS.MATH.7.SP.C.7.B • TEKS 7.6I
GRADE LEVEL BAND:	Grades 6–8 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7, pp. 166–172
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 5 (Side B Weighted Dial) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 2: LAB INVESTIGATION: THE CARNIVAL SIMULATION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Calculate theoretical odds: A = 180/360 = 50%; B = 90/360 = 25%; C = 54/360 = 15%; D = 36/360 = 10%.

Step 2: Spin the arrow 40 times. Tally how many times it lands in Zone A, B, C, and D.

Step 3: Compare: Did Zone A land roughly twice as often as Zone B? Did Zone D land least often?

Step 4: Conclude: The probability of landing in any sector is DIRECTLY proportional to its central angle.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Angular Invariant: Sector Angle / 360° = Theoretical Probability (180° = 50%, 90° = 25%)

PHYSICAL APPARATUS MANIFEST:

Modular Spinner Base + Side B Weighted Dial (Sector A: 180°,
B: 90°, C: 54°, D: 36°)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: DESIGNING A CARNIVAL GAME

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

If it costs $1 to spin: Landing on D wins $5. Landing on C wins $2. Landing on A or B wins nothing.

FORM A MASTER STATION TASKS (Complete All 3):

1 **Calculate expected payout: P(D)×$5 + P(C)×$2.**
Expected Physical Finding: **(0.10 × $5) + (0.15 × $2) = $0.50 + $0.30 = $0.80**

2 **If players pay $1.00 and receive $0.80 on average, who wins long term?**
Expected Physical Finding: **The carnival owner makes $0.20 per spin!**

3 **Is this a fair game?**
Expected Physical Finding: **No, it is mathematically biased toward the house.**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: **[180] [/] [360] -> 0.50 (50% Zone)**

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
180	X: 180	Sector A arc degrees
360	Y: 180 | X: 360	Full circle degrees
/	X: 0.50	Sector probability 50%

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: The Law of Large Numbers

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why are small surveys unreliable in medicine or politics?'
- Ask: 'How does equal-volume pip displacement keep the die from landing on 6 too often?'
- Ask: 'What would happen if we rolled 1,000,000 times?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking the die 'tries' to balance itself out. Emphasize that large numbers work through averaging, not through magical compensation.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A pollster surveys 10 voters and 7 support Candidate A. Why would a statistician say this sample is too small?"
""

OFFICIAL SOLUTION KEY:

A small sample has enormous variance (margin of error ±30%). At least 500–1,000 voters are needed for accuracy.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

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PROB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: The Law of Large Numbers

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does flipping a coin 4 times sometimes give 4 heads, but flipping it 1,000 times almost never gives 1,000 heads? In this lab, observe the Law of Large Numbe

STEP-BY-STEP APPARATUS SETUP:

1. Take one 18mm precision die from your kit.
2. Notice the equal-volume debossed pips: Face 6 uses smaller holes than Face 1 so all faces have equal weight!
3. Prepare to track the appearance of rolling a '6'.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Four 18mm Precision Fair Dice + Dice Landing Tray

DATUM TARGET: Convergence Invariant: Lim(n->∞) |f_n − P| = 0 (Relative frequency converges to theoretical truth)

CALCULATOR KEY: [1] [/] [6] -> 0.1667 (16.67% per face)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
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BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7, Module 3 (Sample Size & Predictions)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 10, Sec. 10.3 • CCSS.MATH.7.SP.C.6 • TEKS 7.6D
GRADE LEVEL BAND:	Grades 6–8 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 9 ('Statistics & Curve Fitting'), pp. 142–156
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 6 (Aha! Discovery: Law of Large Numbers)

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2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: SAMPLE SIZE TRACKING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Trial 1 (Small Sample): Roll the die 6 times. Did 6 appear exactly once? (Often 0 or 2 times).

Step 2: Trial 2 (Medium Sample): Roll 30 times. Count total 6s. Calculate percentage (expected ~16.7%).

Step 3: Trial 3 (Large Class Aggregate): Pool data from all 6 tables in the classroom (total 180 rolls).

Step 4: Notice: As the number of rolls grows from 6 -> 30 -> 180, the percentage of 6s locks onto 16.7%!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Convergence Invariant: Lim(n->∞) |f_n - P| = 0 (Relative frequency converges to theoretical truth)

PHYSICAL APPARATUS MANIFEST:

Four 18mm Precision Fair Dice + Dice Landing Tray

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE ERROR MARGIN SQUEEZE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Look at how the error between experimental and theoretical shrinks.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Theoretical rate for rolling a 6.**
   Expected Physical Finding: **1/6 = 16.67%**

2. **In 6 rolls, getting two 6s gives what percentage?**
   Expected Physical Finding: **2/6 = 33.3% (Large error of +16.7%)**

3. **In 600 rolls, getting 105 6s gives what percentage?**
   Expected Physical Finding: **105/600 = 17.5% (Small error of only +0.8%)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [/] [6] -> 0.1667 (16.67% per face)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Single face
6	Y: 1 | X: 6	Total fair die faces
/	X: 0.1667	Theoretical probability = 16.67%

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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PROB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 4: The 2d6 Triangular Sum Distribution

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is sum 7 six times more likely than sum 2?'
- Ask: 'Why do board game designers put robber/penalty tiles on 7?'
- Ask: 'How does rolling three dice (3d6) turn this triangle into a smooth bell curve?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 2 and 7 have equal chances because they are both numbers on the chart. Remind them: 7 has 6 combinations, but 2 has only 1!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"What is the probability of rolling a sum GREATER than 9 on two dice?"
""

OFFICIAL SOLUTION KEY:

Sums 10 (3 ways), 11 (2 ways), 12 (1 way). Total = 3 + 2 + 1 = 6 ways -> 6/36 = 1/6.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/dice-and-spinner/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PROB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 1: The 2d6 Triangular Sum Distribution

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is 7 the most famous number in board games like Settlers of Catan or Craps? In this lab, discover why adding two fair dice creates a triangular bell curve.

STEP-BY-STEP APPARATUS SETUP:

1. Take two different colored dice: Die A (Teal) and Die B (Amber).
2. List all possible sums: Minimum sum is 1+1 = 2; Maximum sum is 6+6 = 12.
3. Prepare a dot plot chart for sums 2 through 12.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Two 18mm Fair Dice (Teal & Amber) + Sum Tracking Sheet

DATUM TARGET: Combinatoric Peak: 36 Outward Outcomes -> Sum 7 Has 6 Ways (P=6/36=1/6), Sum 2 Has Only 1 Way (P=1/36)

CALCULATOR KEY: [6] [/] [36] -> 0.1667 (16.67% for Sum 7)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 12, Lesson 12.4 (Compound Events)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7, Module 4 (Dice Sums & Bell Curves)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 10, Sec. 10.4 • CCSS.MATH.7.SP.C.8 • TEKS 7.6I
GRADE LEVEL BAND:	Grades 6–8+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7, pp. 170–178
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 2 (2d6 Distribution Table) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 2: LAB INVESTIGATION: THE 36-COMBINATION MAP

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: How many ways can you roll a sum of 2? Only ONE way: (1, 1). P(2) = 1/36.

Step 2: How many ways to roll a sum of 3? TWO ways: (1, 2) and (2, 1). P(3) = 2/36.

Step 3: Count ways to roll 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). Exactly SIX ways! P(7) = 6/36 = 1/6.

Step 4: Roll the two dice 36 times and plot the sums. Look at the shape: It forms a perfect triangular peak centered at 7!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Combinatoric Peak: 36 Outward Outcomes -> Sum 7 Has 6 Ways (P=6/36=1/6), Sum 2 Has Only 1 Way (P=1/36)

PHYSICAL APPARATUS MANIFEST:

Two 18mm Fair Dice (Teal & Amber) + Sum Tracking Sheet

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 6–8+

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE PROBABILITY SYMMETRY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Observe the symmetric drop on both sides of 7.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Compare P(6) and P(8): How many ways to make each?**
   Expected Physical Finding: **Both have exactly 5 ways (5/36 each)**

2. **Compare P(5) and P(9): How many ways to make each?**
   Expected Physical Finding: **Both have exactly 4 ways (4/36 each)**

3. **Compare P(2) and P(12): How many ways to make each?**
   Expected Physical Finding: **Both have only 1 way (1/36 each)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [6] [/] [36] -> 0.1667 (16.67% for Sum 7)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
6	X: 6	Permutations totaling 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1)
36	Y: 6 | X: 36	Total 6×6 Cartesian sample space
/	X: 0.1667	Peak probability for sum 7 = 1/6

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 6–8+

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 4: Expected Value & Game Design

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Can you ever win $3.50 on a single roll of a die?'
- Ask: 'Why is expected value an average over time rather than a possible single outcome?'
- Ask: 'How do state lotteries use expected value to guarantee millions in state revenue?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking $3.50 means you roll 3 on one turn and 4 on the next. Expected value is a weighted long-term mathematical limit.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A coin game pays $4 for heads and $0 for tails. What is the fair entry fee?"
""

OFFICIAL SOLUTION KEY:

(0.5 × $4) + (0.5 × $0) = $2.00.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PROB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 1: Expected Value & Game Design

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What makes a game 'fair'? If a game is fair, neither the player nor the house makes money in the long run. In this lab, calculate the exact mathematical fair pri

STEP-BY-STEP APPARATUS SETUP:

1. Consider this dice game: Roll one fair 18mm die.
2. The house pays you the dollar amount showing on the die: Roll 1 -> Win $1. Roll 6 -> Win $6.
3. Question: What is a fair price to charge a player to enter this game?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Two 18mm Precision Dice + SC-32 Hardware Stack

DATUM TARGET: Expectation Formula: E(X) = Sum(x_i · P(x_i)) -> Break-even Price for Fair Games

CALCULATOR KEY: [7] [ENTER] [3.5] [*] -> E(X) Calculation

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 12, Lesson 12.5 (Expected Value)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7 (Game Theory & Fair Value)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 2, Ch. 10 • CCSS.MATH.HSS.MD.A.2 • TEKS A2.8D
GRADE LEVEL BAND:	Grades 7–9 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 9, pp. 156–164
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 6 (Expected Value Invariant) & Page 7 (RPN TVM)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: CALCULATING EXPECTED VALUE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Every outcome {1, 2, 3, 4, 5, 6} has equal probability 1/6.

Step 2: Calculate average outcome: (1 + 2 + 3 + 4 + 5 + 6) / 6 = 21 / 6 = $3.50.

Step 3: Conclude: Over 100 rolls, the player expects to win $350 ($3.50 per roll on average).

Step 4: If the host charges $3.50 to play, the game is FAIR. If the host charges $4.00, the host profits $0.50 per game!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Expectation Formula: E(X) = Sum(x_i · P(x_i)) -> Break-even Price for Fair Games

PHYSICAL APPARATUS MANIFEST:

Two 18mm Precision Dice + SC-32 Hardware Stack

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: A DOUBLE-DICE CASINO CHALLENGE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

A game pays $10 if you roll a sum of 7 on 2d6. It pays nothing for any other sum.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Probability of rolling a 7.**
   Expected Physical Finding: 6/36 = 1/6

2. **Calculate expected payout: (1/6) × $10.**
   Expected Physical Finding: $10 / 6 = $1.67

3. **If entry costs $2.00, who has the advantage and by how much?**
   Expected Physical Finding: The casino profits by $2.00 – $1.67 = $0.33 per play

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [7] [ENTER] [3.5] [*] -> E(X) Calculation

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3.5	X: 3.5	Average value of a single 6-sided die
ENTER	Y: 3.5 | X: 3.5	Stack lift
2	Y: 3.5 | X: 2	Number of dice
*	X: 7.0	Expected sum of 2d6 = 7.00

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Sample Space Trees & RPN Statistics

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is it usually easier to calculate the probability of an event NOT happening, then subtract from 1?'
- Ask: 'How does the SC-32 y^x function make evaluating large sample spaces instant?'
- Ask: 'Why are casinos so certain of their profit even when players occasionally win big?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding probabilities instead of multiplying for independent events (e.g., doing 1/6 + 1/6 instead of 1/6 × 1/6). Remind them: every branch MULTIPLIES!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If you flip 10 coins, how many total possible outcomes exist in the sample space?"
""

OFFICIAL SOLUTION KEY:

2^{10} = 1,024 outcomes.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PROB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Sample Space Trees & RPN Statistics

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What are the odds of rolling 'Yahtzee' (four matching dice on a single roll)? In this lab, use sample space tree diagrams and the SC-32 power key to solve comple

STEP-BY-STEP APPARATUS SETUP:

1. Take all four precision dice from your kit.
2. Roll all 4 dice simultaneously.
3. Record whether you get: all different numbers, one pair, two pairs, three of a kind, or four of a kind.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Four 18mm Precision Dice + SC-32 Scientific Hardware Calculator

DATUM TARGET: Combinatoric Sample Space: Total Outcomes = 6^k (6^1=6, 6^2=36, 6^3=216, 6^4=1,296)

CALCULATOR KEY: [6] [ENTER] [4] [y^x] -> 1,296 (4d6 Sample Space)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 12 (Permutations & Combinations)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7 (Culminating Probability Systems)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 2, Ch. 10 • CCSS.MATH.HSS.CP.B.9 • TEKS A2.8E
GRADE LEVEL BAND:	Grades 7–10+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 9 ('Statistics & Curve Fitting'), pp. 164–178
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 7 (RPN Statistics Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

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2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: EXPONENTIAL SAMPLE SPACES

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: One die: 6 outcomes. Two dice: 6 × 6 = 36 outcomes.

Step 2: Three dice: 6 × 6 × 6 = 216 outcomes. Four dice: 6^4 = 1,296 total outcomes!

Step 3: Calculate odds of rolling four 1s: Exactly 1 outcome out of 1,296 = 1/1296 (0.077%).

Step 4: Calculate odds of rolling ANY matching four of a kind: 6 outcomes (all 1s, all 2s... all 6s) / 1,296 = 6/1296 = 1/216 (0.46%).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Combinatoric Sample Space: Total Outcomes = 6^k (6^1=6, 6^2=36, 6^3=216, 6^4=1,296)

PHYSICAL APPARATUS MANIFEST:

Four 18mm Precision Dice + SC-32 Scientific Hardware Calcula
tor

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 7–10+

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE BIRTHDAY PARADOX EXTENSION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

What is the probability of rolling 4 dice and having NO TWO DICE MATCH (all four different)?

FORM A MASTER STATION TASKS (Complete All 3):

1 **Die 1: 6 choices. Die 2: 5 choices. Die 3: 4 choices. Die 4: 3 choices.**
Expected Physical Finding: **6 × 5 × 4 × 3 = 360 favorable outcomes**

2 **Divide by total outcomes: 360 / 1296.**
Expected Physical Finding: **360 / 1296 = 5/18 ≈ 27.78%**

3 **What is the probability that AT LEAST TWO dice match?**
Expected Physical Finding: **Complement: 1 − 27.78% = 72.22% (Over 72% chance of a match!)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: **[6] [ENTER] [4] [y^x] -> 1,296 (4d6 Sample Space)**

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
6	X: 6	Single die outcomes
ENTER	Y: 6 | X: 6	Stack lift
4	Y: 6 | X: 4	Number of independent dice
y^x	X: 1,296	Exponential sample space volume: 6^4 = 1,296

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack
spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 7–10+

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